ISTA 130 Python Skills Worksheet

Skill: `if` statements. Write a function called `get_class` that takes one argument, an integer representing the cumulative number of units a student has earned. Return the student's class (`"Freshman"`, `"Sophomore"`, etc.) using the cutoffs 30, 60, 90.

Skills: traversing by element, updating a variable (in this case, building a string). Write a function called `uppers` that takes a string as its sole argument. Return a new string consisting of all of the uppercase letters in the string (hint: `isupper`).

Skills: traversing by index, updating a variable (building a list). Write a function called `add_lists` that takes two lists of equal length and return a new list containing the element-by-element sum of the lists.

Skill: `while` loops. Given the following lines of code, write a while loop that continues rolling two dice until they have different values showing:

```
import random
die1 = random.randint(1, 6)
die2 = random.randint(1, 6)
```

Skills: manipulating sequence types. Given the string `" I heart Rocket  !!!"` stored in variable `astring`, write a line of code that creates the string `'I$heart$Rocket$!!!'`. Write a line of code that grabs the word `"heart"` from `astring`. Write a line of code that creates the string `'  er okt ! '` from the variable. Write a few lines of code that creates the list `[' I HEAR', ' ROCKE', '  !!! ']`.

Skills: file I/O, dictionaries. Write a function called `letter_dict` that takes a filename as an argument. Return a case-insensitive dictionary that maps letters to lists of words that start with those letters, no duplicates. For instance, if the file contains:

```
I heart Rocket.  I HEART ROCKET.
I really, really heart Rocket.  He is radical.
```

Return the dictionary:

```
{
"i": ["i", "is"],
"h": ["heart", "he"],
"r": ["rocket.", "really,", "really", "radical."]
}
```

Create a class called `Rectangle`. Each `Rectangle` object has two instance variables, `width` and `height`. The initializer has parameters corresponding to these variables. `Rectangle` has instance methods called `area` and `perimeter` that take no arguments and return the area and perimeter of the `Rectangle`, respectively. An example of how a Rectangle prints:

```
Width: 4
Height: 6
Area: 24
Perimeter: 20
```